ther pump will probably have a sufficient ow rate to enable the engine to be run at uising speed, but you will need to keep a ose eye on the temperature.

ig a funnel well above the engine and lumb a hose from it to the heat exchanger. Fill it from a bucket. The flow rate will be slow because this is gravity fed, but that's OK because it's going to be hard work keeping the funnel filled! You can adjust the flow rate by moving the funnel up and down. Maintain the engine speed at whatever speed keeps the temperature from going much above normal. I know a couple of people who got themselves out of a tight spot in this manner.

Freshwater side. If the raw-water circuit is functioning as normal and the engine is freshwater cooled, check the level in the coolant recovery bottle (if fitted) or expansion tank. *Warning: Never remove the cap when it's hot. Serious burns may result.* If the level is low, find out where the water is going (adding red food dye may help in tracing leaks). Possibilities are leaking hose connections, heat exchanger or oil cooler cooling tubes that have corroded through (see the Water in the Crankcase section later in this chapter), or a blown head gasket. A blown head gasket will likely cause air bubbles in the cooling system when the engine is running, and these will be visible in the header tank; the header tank may also smell of exhaust fumes. Finally, note that a pressure cap on the header tank with too low a pressure rating, or a pressure cap that has failed, will allow the coolant to boil away over time (the pressure rating will be stamped on the cap; the required pressure is given in the engine specifications).

Troubleshooting Chart 9-3. Overheating on Start-Up

Is water coming from the raw-water discharge? NO	YES → Check the coolant level in the freshwater circuit (if fitted). Caution: Do not remove header-tank pressure cap when hot. If the level is low, refill and find the leak.
Is the raw-water seacock closed? NO	YES → Open and then check the raw-water overboard discharge. The raw-water pump may have failed from running dry.
Is the raw-water strainer plugged? NO	YES → Clean and then check the raw-water discharge as above.
Has the raw-water pump failed? Inspect the drive belt and tension or replace as necessary. Make sure any clutch is operative. If the belt and clutch (if fitted) are OK, remove the pump cover and inspect the impeller vanes. Make sure the impeller turns when the engine turns. NO	YES → Tighten or replace the drive belt as necessary. Replace a damaged impeller. Track down any missing vanes.
Check for collapsed or kinked raw-water hoses, an obstruction over the raw-water inlet on the outside of the hull (break a below-the-waterline hose loose as close to the raw-water seacock as possible and see if there is a good flow into the boat), or a plugged raw-water injection nozzle into the exhaust. Is the water-lift silencer frozen?	

Overheating during normal operation. *Check the oil level.* If a low oil level is causing the engine to overheat, expensive damage may be in the making.

Raw-water side. If the raw-water intake is not set low enough in the hull, a well-heeled sailboat can suck in air. The heeling will have to be kept down until the through-hull can be moved.

The raw-water inlet screen (if fitted) on the outside of the hull may be blocked with a piece of plastic. Throttle down, put the boat in reverse, throttle up, and then shut down the engine once the boat has reverse way on it. With a little luck, the reverse propeller thrust and water flow will wash the plastic clear. To confirm this, loosen a hose below the waterline and see if water flows into the boat. If the flow is restricted, check for barnacles on the inlet screen or in the water intake.

Check the raw-water filter. If it contains a lot of silt, the heat exchanger (or the engine itself on raw-water-cooled boats) may be silted up. Feel the freshwater inlet and outlet pipes to the heat exchanger; if it is doing its job, there should be a noticeable fall in the temperature of the water leaving the heat exchanger. Many heat exchangers have removable end caps and can be rodded out with a suitable wooden dowel. With severe cases of scaling, you will need to obtain special solvents from an engine rebuilding facility and pump these through the heat exchanger; you should get professional advice. Note that if any other heat exchanger or a refrigeration condenser is fitted in series with the engine-cooling circuit, an obstruction in any of these will reduce the raw-water flow through all of them.

Raw-water-cooled engines (without a freshwater circuit) are likely to develop scale around the cylinders over time (Figure 9-38), especially if run above 160°F/71°C. As with heat exchangers, descaling requires professional advice.

Check for collapsed cooling hoses, a loose raw-water pump drive belt, or poor pump performance. This is another reason that a high-set overboard discharge is useful—you can readily measure the pump flow.